

Iterative Calculation (Two Views)

$$Q^n(a) = Q^{(n-1)}(a) + \frac{1}{n} (R^{(n)}(a) - Q^{(n-1)}(a))$$

$Q^{(n-1)}(a) + \alpha_n (R^{(n)}(a) - Q^{(n-1)}(a))$
 Error in $Q^{(n-1)}(a)$
 w.r.t our current
 value $R^{(n)}(a)$, which is
 an instantaneous (single sample)
 estimate of $Q^*(a)$.
 Step-size parameter $\rightarrow 0$

Sign of the
 error
 gives the
 direction
 in which
 we must
 move
 $Q^{(n-1)}(a)$.

We are reducing/increasing $Q^{(n-1)}(a)$, depending on the
 direction of the update. Given the direction, the amount of change
 in $Q^{(n-1)}(a)$ is a function of α_n & the magnitude of the error.

Mean Estimates For Non-Stationary Settings

Exponential Recency Weighted Average

$$Q^n(a) = Q^{(n-1)}(a) + \alpha_n (R^{(n)}(a) - Q^{(n-1)}(a))$$

↓

Initial Bias in Estimates

Does $Q_1(a), \theta_1$ impact/bias your estimates at a later time/attempt?

— Sample mean isn't impacted.

$$Q^{(n)}(a) = \frac{(n-1)}{n} Q^{(n-1)}(a) + \frac{R^{(n)}(a)}{n}$$

For when the first sample value for an a is rec'd, the sample mean is reset to the first sample value.

$$Q^{(n)}(a) = (1-\alpha_n) Q^{(n-1)}(a) + \alpha_n R^{(n)}(a)$$

$$Q^{(2)}(a) = (1-\alpha) Q^{(1)}(a) + \alpha R^{(2)}(a)$$

$$Q^{(3)}(a) = (1-\alpha) \underbrace{Q^{(2)}(a)} + \alpha R^{(3)}(a)$$

$$= (1-\alpha)^2 Q^{(1)}(a) + (1-\alpha)\alpha R^{(2)}(a) + \alpha R^{(3)}(a)$$

Convergence Conditions

$$\sum \alpha_n = \infty$$

$$\sum \alpha_n^2 < \infty$$

- What estimates satisfy them? [~~Go to Classroom~~]

Use Initial Values to Encourage Exploration

- You are using a greedy policy
- How would you select initial values for each action such that all actions get chosen at least once initially?

Optimistic Initial Values

- Exploration!

Upper Confidence Bound Selection

- Can we be smarter when *not* greedy?

$$M_n(x) = \frac{1}{n} \sum_{i=1}^n x_i$$

$$\text{Var}[M_n(x)] = \frac{1}{n^2} \text{Var}\left[\sum_{i=1}^n x_i\right]$$

$$= \frac{1}{n^2} n \text{Var}(x)$$

- Estimates are noisy
 - How does the estimation error change with the number of times an action is selected?

$$E \left[\underbrace{\left(\underbrace{Q^n(a)}_{\text{Sample mean}} - \underbrace{Q^*(a)}_{E[Q^n(a)]} \right)^2} \right] = \text{Var}[Q^n(a)]$$

$$= \frac{\text{Var}[R_t | A_t = a]}{n}$$

Upper Confidence Bound:

Use upper bounds on your estimate $Q_t(a)$, for each a .

Upper Bound for arm $a = Q_t(a) + \text{Absolute Error in the estimate.}$

Don't know the error, of course!

UCB at t :

choose $\arg \max_{a \in A} \left[Q_t(a) + \underbrace{\left(\frac{1}{\sqrt{N_t(a)}} \right)}_{\substack{\downarrow \\ \frac{1}{\sqrt{N_t(a)/\ln(t)}}}} \sqrt{\ln(t)} \right]$

Another well known algo for the bandit setting is
Thompson Sampling.

ϵ -greedy vs. UCB

- UCB does *better*
 - *More on this in HW*
- However, difficult to extend to the more general RL problem
 - Large state spaces, approximations

Gradient Bandit Algorithms

- So far we ...

- (1) Update values $Q_t(a)$ for every arm a .
- (2) Use $Q_t(a)$ to construct a policy, using which we selected an action

Gradient Bandit Algorithms

- Use parameterized PMF over actions
- PMF is a function of an action preference function $H_t(a)$, $a \in A$.
 - A certain evaluation of an action

PMF is parametrized by $H_t(a)$.
- Update evaluation based on obtained rewards
 - This updates the PMF
- An instance of stochastic gradient descent

Gradient Bandit Algorithms

- Use parameterized PMF over actions

$\pi_t(a) = P[A_t = a]$, $a \in \mathcal{A}$, is the PMF at step t .

Write $E[R_t]$ in terms of $\pi_t(a)$.

$$E[R_t] = ? \quad \sum_a E[R_t | A_t = a] \pi_t(a) = \sum_a \phi^*(a) \pi_t(a)$$

Gradient Bandit Algorithms

- Use parameterized PMF over actions

$\pi_t(a) = P[A_t = a]$, $a \in \mathcal{A}$, is the PMF at step t .

We want to maximize
 $E[R_t]$ over the space of policy PMFs.

→ Parameterize the policy PMF

→ Optimize over the space of parameters.

Parameters: $\Pi_t(a)$, $a \in A$.

PMF (softmax)

$$\Pi_t(a) = \frac{e^{\Pi_t(a)}}{\sum_{b \in A} e^{\Pi_t(b)}}, \quad a \in A$$

Goal is to maximize

$E[R_t]$ over the

parameters $\Pi_t(a)$, $a \in A$.

We know how to maximize $f(\theta)$ over the parameter vector $\theta_{1 \times 1}$

First order necessary condition:

$$\nabla f(\theta) = 0 \text{ at } \theta^*$$

$$\text{Let } g(\theta) = \nabla f(\theta)$$

We want to solve $g(\theta) = 0$.

Start with a guess θ_0 . In the neighborhood of θ_0 , the following is a good approximation:

$$g(\theta) \approx g(\theta_0) + (\theta - \theta_0)^T \nabla g(\theta_0)$$

$$g(\theta) \approx g(\theta_0) + \overbrace{\nabla g(\theta_0)}^{\text{Jacobian}} (\theta - \theta_0)_{1 \times 1}$$

How do we get the above?

What do we do next?

Setting $g(\theta) = 0$, gives us:

$$0 = g(\theta_0) + \nabla g(\theta_0) (\theta - \theta_0)$$

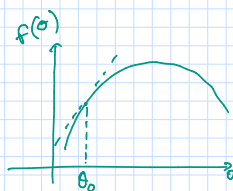
$$\nabla g(\theta_0) \theta = -g(\theta_0) + \nabla g(\theta_0) \theta_0$$

$$\theta_1 = -\nabla g(\theta_0)^{-1} g(\theta_0) + \theta_0$$

$$= \theta_0 - \nabla g(\theta_0)^{-1} g(\theta_0)$$

$$= \theta_0 - [\nabla^2 f(\theta_0)]^{-1} g(\theta_0)$$

$$= \theta_0 - [\nabla^2 f(\theta_0)]^{-1} \nabla f(\theta_0)$$



GRADIENT ASCENT

$$\theta_1 = \theta_0 + \alpha \nabla f(\theta_0)$$

$$\theta_2 = \theta_1 + \alpha \nabla f(\theta_1)$$

⋮

Back to our problem

$$\text{maximize } E[R_t] \\ (H_t(1), H_t(2), \dots, H_t(k))$$

We want to solve

$$\nabla_{\bar{H}_t(\alpha)} E[R_t] = 0$$

Just do:

$$\bar{H}_{t,1}(\alpha) = \bar{H}_{t,0}(\alpha) + \alpha \nabla E[R_t] \\ \vdots \\ \vdots$$

Is there a problem?